

Mohd Bilal Khan

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EDUCATION

Indian Institute of Information Technology (IIIT), Pune

Bachelor of Technology in Electronics and Communication Engineering

Pune, India

Aug 2023 – May 2027

TECHNICAL SKILLS

Languages: C++, Python, Go (Golang), Java, JavaScript, Shell Scripting

AI/ML Frameworks: PyTorch, LangGraph, RAG, CLIP, Transformers

Web & DevOps: React, Flask, Streamlit, Docker, Git, GitHub, Postman, Selenium

Databases: Neo4j (Graph DB), ChromaDB, FAISS, Supabase

PROFESSIONAL EXPERIENCE

Student Researcher & Team Lead

Oct 2025 – Present

Digital Forensics Lab, IIIT Pune

Pune, India

- Built an AI-powered digital forensics platform with **6 ML models (YOLOv8, BERT, BART, Whisper, Random Forest, SigLIP)** orchestrated via a **FastAPI** backend serving 7 API router groups.
- Developed NLP evidence extraction pipeline processing documents through **NER, topic modeling, keyword scanning, and summarization** to detect threats across **8 criminal categories**.
- Designed a **React/TypeScript** forensic dashboard with real-time analysis monitoring, automated threat verdicts, and one-click **PDF/JSON** report generation.
- Implemented ML-based malware detection using **PE header feature extraction, YARA rule matching**, and Docker-sandboxed binary analysis.
- Engineered an async job orchestration system processing evidence files in parallel with **remote GPU inference** support and progress tracking via polling.

KEY PROJECTS

Sanjaya AI: Agentic Pharma Innovation Platform | [Live Demo](#) | *LangGraph, FastAPI, React, Supabase*

- Engineered an end-to-end **Agentic AI system** to automate pharmaceutical molecule research, reducing discovery timelines from **2–3 months to minutes**.
- Orchestrated a master-worker architecture using **LangGraph** and **OpenAI SDK**, where specialized agents autonomously conduct patent landscaping, clinical trial analysis, and market intelligence gathering.
- Built a responsive **React/Vite** interface and **FastAPI** backend that synthesizes multi-source data into comprehensive PDF/Excel reports, featuring secure internal document ingestion.

Sentinel Core: AI-Powered Digital Forensics Analysis Platform | *Python, FastAPI, React, TypeScript, PyTorch, Docker*

- Designed and built a full-stack AI-powered digital forensics platform enabling investigators to analyze USB drives and file systems through **6 specialized ML pipelines** with automated threat classification (**CLEAN / SUSPICIOUS / MALICIOUS**).
- Engineered **5 AI analysis pipelines** — Vision (**YOLOv8** object detection + violence classification), Text/NLP (**BERT** NER + **BART** summarization), Malware (Random Forest PE analysis + TF-IDF URL classification), File Integrity (**YARA** rule scanning), and Audio (**Whisper** speech-to-text).
- Built a **Master Agent orchestrator** leveraging **GPT-4** and **Google Gemini** to cross-correlate findings across pipelines and generate executive, technical, and legal case summaries.
- Implemented a real-time job orchestration system using **FastAPI BackgroundTasks** with **ThreadPoolExecutor** parallelism (8 workers), UUID-based job tracking, and client-side progress polling.
- Developed a professional **PDF forensic report generator** using ReportLab with verdict banners, risk matrices, and per-file evidence detail cards for law enforcement use.

ZeroTrace: High-Performance Data Erasure Utility | *Go, Shell Scripting*

- Engineered a professional-grade security tool for Windows, Linux, macOS, and Android to perform irreversible data sanitization using **NIST SP 800-88** and DoD 5220.22-M protocols.
- Achieved a complete drive wipe on a **512GB NVMe SSD in 5.5 hours**, optimizing I/O operations for speed.
- Developed specifically for a national hackathon sponsored by the Government of India.

ACHIEVEMENTS & LEADERSHIP

National Hackathon Winner: Secured **Rank 130** (Top 1.3%) out of 10,000+ teams in the HackRx national hackathon.

Academic Excellence: Achieved a **Top 2% rank** (among 1.1 million candidates) in the JEE Mains examination.

Research Publication: Authored a manuscript (in preparation) on Chili Plant Disease Classification using CNNs, achieving **92.6% accuracy**.

Leadership: Selected as **Google DSC Campus Ambassador** for IIIT Pune; Active contributor to GirlScript Summer of Code (2025).

Athletics: Silver Medalist at the Inter-IIIT National Football Tournament.